

Addendum 2 — Quality-First Revision

Prepared 2026-08-01 · Supersedes: Addendum 1 §2 and §7

1. The optimization target moves

With wall-clock time removed as a constraint, throughput stops being a design consideration at all. The correct metric becomes edits avoided per page — how much correction the certifying translator does not have to make. Any amount of additional computation is worth spending if it removes even a handful of corrections from each page, because the translator’s time is now the only scarce resource in the project.

This inverts several earlier recommendations. It also opens techniques that were previously ruled out on cost grounds, and those are worth more than the quantization decision that dominated the last revision.

2. Quantization: revert to Q8_0

The reasoning that produced the Q5_K_M recommendation was purely a throughput trade. Without that trade, the argument runs entirely the other way.

Quantization loss falls disproportionately on the linguistic patterns a model has seen least — exactly the Slovene morphology this project depends on. Q8_0 is where the loss curve flattens; below it, degradation is measurable, and for a less-resourced language it is larger than the general benchmarks suggest.

Quant	Size	Speed	Decision
Q8_0	~13 GB	~4 tok/s	Recommended. Effectively lossless; ample memory headroom remains
BF16 (unquantized)	~24 GB	~2 tok/s	Not recommended. Fits, but leaves under 3 GiB for context and desktop. Gain over Q8_0 is marginal; the memory risk is not
Q6_K	9.66 GB	~5–6 tok/s	Now a fallback only
Q5_K_M / Q4_K_M	7–8.5 GB	~6–8 tok/s	Withdrawn. No longer any reason to accept the loss

Worth confirming empirically all the same: translate one representative passage at Q8_0 and at Q5_K_M and have the translator compare. If they cannot tell the difference on legal register, that is useful information about where the real bottleneck lies.

3. Multi-pass translation — the substantive gain

This is where free compute actually buys quality, and it matters considerably more than the quantization choice above. A single generation pass is the weakest possible use of an unconstrained compute budget.

3.1 Back-translation as an omission detector

The failure mode most dangerous in this corpus is the fluent, confident, wrong translation — a dropped negation, an omitted subordinate clause, an inverted attribution. These are invisible on inspection of the target text alone, because the target reads perfectly well.

Back-translation catches them. Take the translated output, translate it back to the source language using a different model, and compare the result against the original source. Meaning-preserving variation produces paraphrase; meaning-destroying errors produce visible divergence. Using a different model for the reverse pass is essential — the same model will tend to reproduce its own misreading and confirm itself.

Flag segments where the back-translation diverges materially from the source, and hand the translator that list. This is the single highest-value application of surplus compute in the entire workflow.

3.2 Self-critique refinement

Present the model with the source, its own draft, and the agreed glossary, and ask it to identify and correct errors rather than to retranslate. This reliably catches glossary violations and register inconsistencies that a first pass misses, and it costs only another pass.

3.3 Proposed pass structure

Pass	Model	Function	Output
1	GaMS3-12B Q8_0	Draft translation with glossary injected	Draft target text
2	GaMS3-12B Q8_0	Self-critique against source and glossary	Revised target text
3	qwen3.6 35B-A3B	Back-translation to source language	Reverse text for comparison
4	Deterministic script	Divergence and consistency checks	Flagged-segment report

Total cost is roughly three to four times a single pass. At Q8_0 that puts a 500-page corpus in the region of four to six days of continuous running — comfortably inside the stated tolerance.

4. Deterministic checks cost nothing — run all of them

Pass 4 above is not a model invocation. It is ordinary code, it runs in seconds, and it catches a category of error that language models are structurally bad at noticing. It should be built regardless of how the rest of the pipeline develops, and it will likely repay more effort per line than anything else here.

Checks worth implementing:

- Every number, date, and monetary amount present in the source appears in the target. Numeric drift is the highest-consequence error class in legal evidence and is trivially detectable by comparison.
- Case numbers, statutory citations, and institution names reproduced verbatim per the agreed policy, not translated.
- Glossary terms rendered with the agreed target term. Any deviation reported with document and segment reference.
- No empty, untranslated, or duplicated segments; no segment where the target is identical to the source unless policy requires it.
- Back-translation divergence above a threshold, from pass 3.
- The same source segment translated inconsistently across different documents — a corpus-level check no per-document review will surface.

The report from these checks is what the translator should receive alongside the drafts. It converts an undifferentiated proofreading obligation into a prioritized worklist, which is precisely the leverage this project needs.

5. Multi-day run hygiene

A batch that runs for days introduces failure modes a short job does not have. Three are worth engineering for before starting.

Checkpointing and resumability

The job must be resumable at segment granularity. A crash at hour sixty must not restart from zero. Write each completed segment to disk immediately, keyed by a stable identifier, and have the driver skip anything already present on restart. This also makes the run auditable, which matters given the material.

Thermal and power

A 15 W base-TDP mobile processor at sustained full load for days is outside its typical duty cycle. Keep the machine on mains power, elevated for airflow, and monitor package temperature. Holding the battery at full charge continuously also accelerates its degradation; if the firmware exposes a charge threshold, cap it for the duration.

```
# monitor while running
watch -n5 'sensors | grep -i package; ollama ps'

# cap battery charge if supported (path varies by model)
cat /sys/class/power_supply/BAT0/charge_control_end_threshold
echo 80 | sudo tee /sys/class/power_supply/BAT0/charge_control_end_threshold
```

Suspend and idle behavior

Confirm the machine will not suspend mid-run. Lid-close and idle-suspend defaults will silently halt a multi-day job.

```
systemd-inhibit --what=idle:sleep:handle-lid-switch \
    --why="translation batch" ./run_batch.sh
```

6. Revised planning arithmetic

At Q8_0 with the four-pass structure, generation cost per scanned page rises roughly fourfold against the single-pass estimate in Addendum 1.

Scanned pages	Single pass	Four-pass structure	Practical schedule
250	~12 hours	~2 days	One weekend
500	~24 hours	~4 days	Under a week
1,000	~48 hours	~8 days	Two weeks with interruptions

Excludes OCR, prefill, and the deterministic checks, which are comparatively cheap. Prefill remains worth optimizing per Addendum 1 §6 even with time abundant, because a stable cached prefix is free to arrange.

7. What does not change

Abundant compute does not relieve any of the human-bound constraints, and it is worth being explicit that the following remain exactly as stated:

- The OCR text layer still requires human verification. Dual-engine cross-checking narrows the work; it does not eliminate it.
- The glossary still requires the translator's decisions before bulk translation begins. No amount of computation substitutes for fixing the target terminology once, up front.
- The professional and procedural questions in Workflow Architecture §9 still gate everything downstream, and remain the correct first action.

- Phase 1 remains the decision point. Three pages, hand-fed, translator's judgment. If machine output is not materially faster to edit than translating fresh, running the machine for a week does not change that verdict — it only makes the wrong answer arrive more expensively.